Assignment #1: Fingerprint Recognition with NBIS

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I. Introduction

We evaluate fingerprint recognition with the NIST Biometric Image Software (NBIS) [1]: we select a matching threshold and classify fingerprint pairs as *same* or *different finger*, first directly and then only within groups of the same pattern type.

II. Methodology

Images are converted from TIFF to PNG, as MINDTCT fails on TIFF. MINDTCT detects minutiae (ridge endings and bifurcations); BOZORTH3 compares two minutiae sets independently of rotation and translation and returns an unnormalised score. All fingerprints are compared with all others; the class is the finger id from the file name. Quality is estimated with NFIQ (1 = best, 5 = worst).

Threshold selection. A pair is classified as *same finger* if its score is at least t. We compute the accuracy for every integer t up to the highest score and keep the best (smallest on ties); checking all values avoids stopping at a local maximum.

Classification. Each unordered pair is compared once (no A vs. A or B vs. A). We report accuracy (correct / all pairs), false acceptance (FAR) and false rejection rate (FRR). In the second experiment, PCASYS first assigns each fingerprint a type (A, L, R, S, T, W); only pairs of the same type are matched, others are classified as *different finger*.

III. Experiments

We use FVC2002 DB1_B [2]: 10 fingers with 8 impressions each (388×374 px, 500 dpi optical sensor), giving 3,160 unique pairs (280 genuine, 2,880 impostor). As instructed, the whole dataset is used for threshold selection and evaluation. NBIS tools use default parameters. PCASYS needs WSQ images of at least 512×480 px, so images are padded with white to 600×600 and compressed with `cwsq` (bitrate 0.75); we use its MLP classifier.

IV. Results and Discussion

A. Results

The best threshold is $t = 27$ (Table I). Impostor scores lie between 0 and 30 (mean 7.3), genuine ones between 0 and 289 (mean 95.5) (Fig. 1).

TABLE I
Classification of the 3,160 unique pairs with the selected $t = 27$. *Matched*: pairs compared with BOZORTH3.

Experiment	Matched	Acc. [%]	FAR [%]	FRR [%]
No types	3,160	98.99	0.10	10.36
PCASYS types	1,172	98.29	0.03	18.93

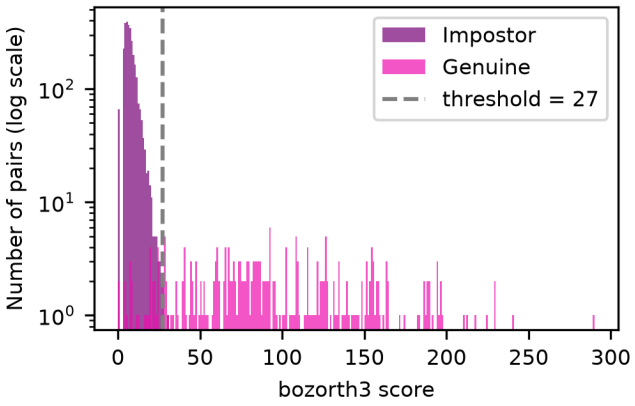


Fig. 1. BOZORTH3 scores of the 280 genuine and 2,880 impostor pairs (log-scale y-axis). The dashed line marks the selected threshold $t = 27$.

In the similarity matrix (Fig. 2) each finger forms a block on the diagonal. NFIQ rates 38 images with quality 1, 35 with 2 and 7 with 3 (Fig. 3).

B. Discussion

The classes are imbalanced: rejecting every pair would already give 91.14% accuracy. Almost all errors are false rejections, linked to low quality: the 7 quality-3 images come from fingers 101, 107 and 109 (four from 109, which has a weak block in Fig. 2).

With PCASYS, four impressions get a different type than the rest of their finger, so 26 genuine pairs are rejected without matching, nearly doubling the FRR. In exchange, two of three false acceptances disappear and 63% of comparisons are avoided.

V. Conclusion

With $t = 27$, chosen by maximising accuracy, pair classification reaches 98.99% accuracy. Matching within PCASYS

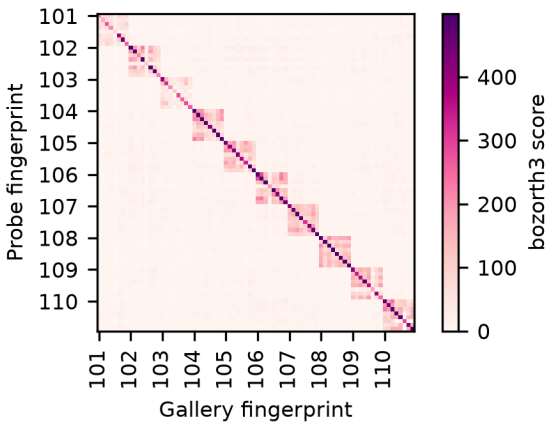


Fig. 2. Similarity matrix of all 80×80 comparisons, sorted by finger. Pale rows inside a block are impressions that match their finger poorly.

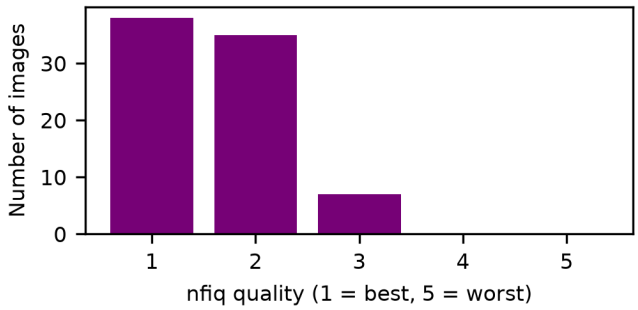


Fig. 3. Images per NFIQ quality level (1 = best, 5 = worst).

types saves 63% of comparisons at a small accuracy cost, as a wrong type cannot be corrected later. As the threshold was tuned and evaluated on the same small dataset, the results are optimistic.

References

[1] C. I. Watson, M. D. Garris, E. Tabassi, C. L. Wilson, R. M. McCabe, S. Janet, and K. Ko, "User's guide to NIST biometric image software (NBIS)," National Institute of Standards and Technology, Tech. Rep. NISTIR 7392, 2007.

[2] D. Maio, D. Maltoni, R. Cappelli, J. L. Wayman, and A. K. Jain, "FVC2002: Second fingerprint verification competition," in *International Conference on Pattern Recognition (ICPR)*, vol. 3. IEEE, 2002, pp. 811–814.